

f) $\forall x \exists y (A(x,y))$: For all students x of my class and for all students y of my class, x has sent an email message to y .
Simple: Each student of my class has sent an email message to all the students of my class.

4) Let $P(x,y)$ be the statement "Student x has taken class y ," where the domain for x consists of all students in your class and for y consists of all computer science courses at your school. Express each of these quantifications in English.

a) $\exists x \exists y P(x,y)$: There exists a student x in my class and there also exists a computer science course y in my school such that Student x has taken class y .

Simple: Some student in my class has taken some computer science course in my school.

b) $\exists x \forall y P(x,y)$: There exists a student x in my class such that for all the computer science courses y of my school, x has taken class y .

Simple: Some student in my class has taken all the computer science courses in my school.

c) $\forall x \exists y P(x,y)$: For all students x of my class, there exists a computer science course y in my school where Student x has taken class y .

Simple: Each student in my class has taken some computer science course in my school.

d) $\exists y \forall x P(x,y)$: There exists a computer science course in my school such that for all the students x of my class, x has taken class y .

Simple: Some computer science course in my school has been taken by every student of my class.

e) $\forall y \exists x P(x,y)$: For all computer science courses in my school, there exists a student x of my class s.t. x has taken class y .

Simple: Each computer science course of my school has been taken by some student of my class.